

Mini-Projet-Système d'exploitation

Groupe :

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1. Installation

Pour ce projet, nous avons utilisé Ubuntu Server installé sur une machine virtuelle VMware. L'installation a été faite en mode texte, sans interface graphique. Une fois le système installé et le terminal accessible, nous avons commencé à exécuter les différentes commandes Linux demandées dans le cadre du projet.

2. Environnement Shell

L'environnement utilisé est le terminal (Shell Bash), qui permet d'exécuter toutes les commandes système

3. Commandes de base

➤ Navigation dans les dossiers

- ls : liste le contenu d'un dossier
- cd : change de dossier
- pwd : affiche le dossier courant
- mkdir : crée un dossier
- touch : crée un fichier vide
- cp : copie un fichier ou un dossier
- mv : déplace ou renomme un fichier ou dossier
- rm : supprime un fichier ou un dossier

```
obunto@obunto-VMware-Virtual-Platform:~$ ls
Cours  Documents  expo_system  Music    Public    snap  Templates  Videos
Desktop Downloads  monrep      Pictures pyProjects TD    TP
obunto@obunto-VMware-Virtual-Platform:~$ cd expo_system
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ pwd
/home/obunto/expo_system
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ mkdir minproj
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ touch exo.txt
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ ls
exo.txt  minproj
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ cp exo.txt exo2.txt
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ ls
exo2.txt  exo.txt  minproj
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ mv exo2.txt home/obunto/expo_system/minproj/
mv: cannot move 'exo2.txt' to 'home/obunto/expo_system/minproj/': No such file or directory
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ mv exo2.txt /home/obunto/expo_system/minproj/
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ ls /home/obunto/expo_system/minproj/
exo2.txt
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ ls
exo.txt  minproj
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ rm exo.txt
```

➤ Fichiers et éditeur

- nano, vim : éditeurs de texte en ligne de commande
- less ,cat, more : affiche le contenu d'un fichier
- head, tail : affiche le début ou la fin d'un fichier

```
GNU nano 7.2                                exo.txt *
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28

[ Read 30 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace   ^U Paste      ^J Justify
^C Location  M-U Undo     M-A Set Mark
^_ Go To Line M-E Redo     M-6 Copy
```

```
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ ls
exo.txt
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ nano exo.txt
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ less exo.txt
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ head exo.txt
1
2
3
4
5
5
6
7
8
9
10
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ tail exo.txt
21
22
23
24
25
26
27
28
29
30
obunto@obunto-VMware-Virtual-Platform:~/expo_system$
```

➤ Gestion des droits et utilisateurs

- chmod : change les permissions
- chown : change le propriétaire
- groups, id, whoami : informations sur l'utilisateur connecté

```
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ ls -l
total 0
-rw-rw-r-- 1 obunto obunto 0 May 14 23:02 exo.txt
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ chmod ugo+wx exo.txt
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ sudo chown user2 exo.txt
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ ls -l
total 0
-rwxrwxrwx 1 user2 obunto 0 May 14 23:02 exo.txt
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ groups
obunto adm cdrom sudo dip plugdev users lpadmin
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ id
uid=1000(obunto) gid=1000(obunto) groups=1000(obunto),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users),114(lpadmin)
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ whoami
obunto
obunto@obunto-VMware-Virtual-Platform:~/expo_system$
```

➤ Redirection et pipes

- > : envoie la sortie d'une commande dans un fichier
- >> : ajoute à la fin d'un fichier
- < : lit à partir d'un fichier
- | : relie deux commandes
- tee : affiche et enregistre en même temps

```
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ touch exo.txt
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ echo "hello" > exo.txt
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ echo "my name is kebir" >> exo.txt
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ sort < exo.txt
hello
my name is kebir
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ less | grep "hello" exo.txt
hello
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ echo "this is our presantation" | tee exo.txt
this is our presantation
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ sort < exo.txt
this is our presantation
obunto@obunto-VMware-Virtual-Platform:~/expo_system$
```

Partie 2 : Administration de base

Gestion des utilisateurs et groupes :

- useradd : crée un utilisateur
- passwd : définit un mot de passe
- userdel : supprime un utilisateur
- groupadd : crée un groupe
- usermod : modifie un utilisateur (ex : ajout à un groupe)
- gpasswd : gère les membres d'un groupe
- chmod, chown : modifient les droits et le propriétaire

```
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ sudo useradd user1
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ sudo passwd user1
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ sudo userdel user2
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ sudo groupadd testgroup
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ sudo usermod -aG testgroup user1
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ groups user1
user1 : user1 testgroup
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ sudo gpasswd -a obunto testgroup
Adding user obunto to group testgroup
obunto@obunto-VMware-Virtual-Platform:~/expo_system$
```

Surveillance du système

➤ Commandes utiles

- top, htop : affichent les processus

```
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ top

top - 02:27:58 up 6:10, 1 user, load average: 0.02, 0.03, 0.05
Tasks: 321 total, 1 running, 320 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1.1 us, 3.0 sy, 0.0 ni, 95.8 id, 0.0 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 3866.7 total, 1376.5 free, 1562.7 used, 1198.5 buff/cache
MiB Swap: 3776.0 total, 3776.0 free, 0.0 used, 2304.0 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
2203	obunto	20	0	4534248	290672	125944	S	3.9	7.3	6:54.78	gnome-shell
1062	root	20	0	762892	19400	16328	S	2.6	0.5	0:08.03	NetworkManager
986	message+	20	0	12176	6996	4436	S	1.6	0.2	0:05.24	dbus-daemon
1410	mysql	20	0	2377400	393444	36408	S	1.3	9.9	5:18.23	mysqld
985	avahi	20	0	8668	4232	3976	S	1.0	0.1	0:02.05	avahi-daemon
2316	obunto	20	0	551856	11216	9936	S	1.0	0.3	0:01.04	gsd-sharing
6592	obunto	20	0	23184	5864	3688	R	1.0	0.1	0:00.34	top
399	root	19	-1	51196	19184	17904	S	0.7	0.5	1:05.37	systemd-journal
2656	obunto	20	0	1382768	24652	21196	S	0.7	0.6	0:04.24	evolution-calen
2786	obunto	20	0	842304	29636	25924	S	0.7	0.7	0:01.10	evolution-addre
6538	obunto	20	0	3154508	60772	46556	S	0.7	1.5	0:00.53	gjs
6594	root	20	0	312460	8808	7912	S	0.7	0.2	0:00.02	nm-dispatcher
1	root	20	0	23172	14144	9408	S	0.3	0.4	0:16.88	systemd
119	root	20	0	0	0	0	I	0.3	0.0	0:00.17	kworker/u512:1-ipv6_addrconf
553	systemd+	20	0	21836	12948	10772	S	0.3	0.3	0:03.87	systemd-resolve
1023	syslog	20	0	222564	6732	4556	S	0.3	0.2	0:01.22	rsyslogd
1937	obunto	20	0	21160	12868	9664	S	0.3	0.3	0:10.76	systemd
1964	obunto	20	0	10988	6648	4608	S	0.3	0.2	0:08.63	dbus-daemon
2529	obunto	20	0	1383732	88824	73208	S	0.3	2.2	0:02.63	evolution-sourc
2534	obunto	20	0	555252	24240	20784	S	0.3	0.6	0:01.05	goa-daemon
2810	obunto	20	0	710392	14104	11588	S	0.3	0.4	0:03.94	xdg-desktop-por
2991	obunto	20	0	566948	56892	43432	S	0.3	1.4	1:10.12	gnome-terminal-
4962	cups-br+	20	0	268464	20252	17308	S	0.3	0.5	0:02.99	cups-browsed
5970	root	20	0	0	0	0	I	0.3	0.0	0:00.50	kworker/2:0-events

- ps aux : liste tous les processus

```
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.3  23172 14144 ?        Ss   May14    0:16 /sbin/init splash
root           2  0.0  0.0      0     0 ?        S    May14    0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    May14    0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   May14    0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<   May14    0:00 [kworker/R-sync_wq]
root           6  0.0  0.0      0     0 ?        I<   May14    0:00 [kworker/R-slub_flushwq]
root           7  0.0  0.0      0     0 ?        I<   May14    0:00 [kworker/R-netns]
root          11  0.0  0.0      0     0 ?        I    May14    0:00 [kworker/u512:0-ipv6_addrconf]
root          12  0.0  0.0      0     0 ?        I<   May14    0:00 [kworker/R-mm_percpu_wq]
root          13  0.0  0.0      0     0 ?        I    May14    0:00 [rcu_tasks_kthread]
root          14  0.0  0.0      0     0 ?        I    May14    0:00 [rcu_tasks_rude_kthread]
root          15  0.0  0.0      0     0 ?        I    May14    0:00 [rcu_tasks_trace_kthread]
root          16  0.0  0.0      0     0 ?        S    May14    0:00 [ksoftirqd/0]
root          17  0.0  0.0      0     0 ?        I    May14    0:04 [rcu_preempt]
root          18  0.0  0.0      0     0 ?        S    May14    0:00 [rcu_exp_par_gp_kthread_worker/1]
root          19  0.0  0.0      0     0 ?        S    May14    0:00 [rcu_exp_gp_kthread_worker]
root          20  0.0  0.0      0     0 ?        S    May14    0:00 [migration/0]
root          21  0.0  0.0      0     0 ?        S    May14    0:00 [idle_inject/0]
root          22  0.0  0.0      0     0 ?        S    May14    0:00 [cpuhp/0]
```

- df -h : affiche l'espace disque utilisé
- du -sh : taille d'un dossier
- uptime : durée de fonctionnement
- free -m : mémoire utilisée

```
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            387M  2.1M  385M   1% /run
/dev/sda2        20G   14G   5.1G  73% /
tmpfs            1.9G     0   1.9G   0% /dev/shm
tmpfs            5.0M   8.0K   5.0M   1% /run/lock
tmpfs            387M  124K  387M   1% /run/user/1000
/dev/sr0         6.0G   6.0G     0 100% /media/obunto/Ubuntu 24.04.2 LTS amd64
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ du -sh /home
du: cannot read directory '/home/user2': Permission denied
3.0G    /home
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ uptime
 02:34:41 up  6:17,  1 user,  load average: 0.11, 0.07, 0.05
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ free -m
              total        used        free        shared  buff/cache        available
Mem:           3866         1585         1325           34         1242         2281
Swap:          3775           0         3775
```


- journalctl : affiche les journaux
- dmesg : messages du noyau

```
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ journalctl
Apr 14 11:56:59 obunto-VMware-Virtual-Platform kernel: Linux version 6.11.0-17-generic (buildd@lcy02-amd64-038) (x86_64>
Apr 14 11:56:59 obunto-VMware-Virtual-Platform kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.11.0-17-generic root=UU>
Apr 14 11:56:59 obunto-VMware-Virtual-Platform kernel: KERNEL supported cpus:
Apr 14 11:56:59 obunto-VMware-Virtual-Platform kernel: Intel GenuineIntel
Apr 14 11:56:59 obunto-VMware-Virtual-Platform kernel: AMD AuthenticAMD
Apr 14 11:56:59 obunto-VMware-Virtual-Platform kernel: Hygon HygonGenuine
Apr 14 11:56:59 obunto-VMware-Virtual-Platform kernel: Centaur CentaurHauls
Apr 14 11:56:59 obunto-VMware-Virtual-Platform kernel: zhaoxin Shanghai
Apr 14 11:56:59 obunto-VMware-Virtual-Platform kernel: Disabled fast string operations
Apr 14 11:56:59 obunto-VMware-Virtual-Platform kernel: BIOS-provided physical RAM map:
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ sudo dmesg
[ 0.000000] Linux version 6.11.0-25-generic (buildd@lcy02-amd64-027) (x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2
~24.04) 13.3.0, GNU ld (GNU Binutils for Ubuntu) 2.42) #25~24.04.1-Ubuntu SMP PREEMPT_DYNAMIC Tue Apr 15 17:20:50 UTC 2
(Ubuntu 6.11.0-25.25~24.04.1-generic 6.11.11)
[ 0.000000] Command line: BOOT_IMAGE=/boot/vmlinuz-6.11.0-25-generic root=UUID=ac830d26-b2b4-4e66-891b-e66d71e30373 r
o quiet splash
[ 0.000000] KERNEL supported cpus:
[ 0.000000] Intel GenuineIntel
[ 0.000000] AMD AuthenticAMD
[ 0.000000] Hygon HygonGenuine
[ 0.000000] Centaur CentaurHauls
[ 0.000000] zhaoxin Shanghai
[ 0.000000] Disabled fast string operations
[ 0.000000] BIOS-provided physical RAM map:
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x00000000000009e7ff] usable
[ 0.000000] BIOS-e820: [mem 0x00000000000009e800-0x00000000000009ffff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000000dc000-0x0000000000000fffff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000100000-0x000000000bfecffff] usable
```

- tail -f /var/log/syslog : suivi des logs en direct

```
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ sudo tail -f /var/log/syslog
2025-05-15T02:46:09.246464+00:00 obunto-VMware-Virtual-Platform systemd-resolved[553]: Using degraded feature set UDP in
stead of TCP for DNS server 192.168.86.2.
2025-05-15T02:46:38.562253+00:00 obunto-VMware-Virtual-Platform systemd[1]: Starting fwupd-refresh.service - Refresh fwu
pd metadata and update motd...
2025-05-15T02:46:38.631374+00:00 obunto-VMware-Virtual-Platform systemd[1]: fwupd-refresh.service: Deactivated successfu
lly.
2025-05-15T02:46:38.631878+00:00 obunto-VMware-Virtual-Platform systemd[1]: Finished fwupd-refresh.service - Refresh fwu
pd metadata and update motd.
2025-05-15T02:46:41.229612+00:00 obunto-VMware-Virtual-Platform systemd-resolved[553]: Using degraded feature set TCP in
stead of UDP for DNS server 192.168.86.2.
2025-05-15T02:46:41.234973+00:00 obunto-VMware-Virtual-Platform systemd-resolved[553]: Using degraded feature set UDP in
stead of TCP for DNS server 192.168.86.2.
2025-05-15T02:46:41.241933+00:00 obunto-VMware-Virtual-Platform systemd-resolved[553]: Using degraded feature set TCP in
stead of UDP for DNS server 192.168.86.2.
2025-05-15T02:46:41.245795+00:00 obunto-VMware-Virtual-Platform systemd-resolved[553]: Using degraded feature set UDP in
stead of TCP for DNS server 192.168.86.2.
2025-05-15T02:47:45.238792+00:00 obunto-VMware-Virtual-Platform systemd-resolved[553]: Using degraded feature set TCP in
stead of UDP for DNS server 192.168.86.2.
2025-05-15T02:47:45.241861+00:00 obunto-VMware-Virtual-Platform systemd-resolved[553]: Using degraded feature set UDP in
stead of TCP for DNS server 192.168.86.2.
```

Processus et services

- systemctl : contrôle des services (start, stop, status)

```
ibtissam@ibtissam-VMware-Virtual-Platform:~$ sudo systemctl start apache2
[sudo] password for ibtissam:
ibtissam@ibtissam-VMware-Virtual-Platform:~$ sudo systemctl stop apache2
ibtissam@ibtissam-VMware-Virtual-Platform:~$ sudo systemctl status apache2
○ apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: inactive (dead) since Thu 2025-05-15 16:41:19 GMT; 23s ago
     Duration: 5min 22.953s
    Docs: https://httpd.apache.org/docs/2.4/
   Process: 6256 ExecStop=/usr/sbin/apachectl graceful-stop (code=exited, status=0/SUCCESS)
    Main PID: 5663 (code=exited, status=0/SUCCESS)
      CPU: 184ms

May 15 16:35:56 ibtissam-VMware-Virtual-Platform systemd[1]: Starting apache2.service - The Apache HTTP Server...
May 15 16:35:56 ibtissam-VMware-Virtual-Platform apachectl[5662]: AH00558: apache2: Could not reliably determine the se>
May 15 16:35:56 ibtissam-VMware-Virtual-Platform systemd[1]: Started apache2.service - The Apache HTTP Server.
May 15 16:41:19 ibtissam-VMware-Virtual-Platform systemd[1]: Stopping apache2.service - The Apache HTTP Server...
May 15 16:41:19 ibtissam-VMware-Virtual-Platform apachectl[6258]: AH00558: apache2: Could not reliably determine the se>
May 15 16:41:19 ibtissam-VMware-Virtual-Platform systemd[1]: apache2.service: Deactivated successfully.
May 15 16:41:19 ibtissam-VMware-Virtual-Platform systemd[1]: Stopped apache2.service - The Apache HTTP Server.
lines 1-16/16 (END)
```


- service : gestion ancienne des services

```
ibttissam@ibttissam-VMware-Virtual-Platform:~$ sudo service apache2 status
[sudo] password for ibttissam:
○ apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: inactive (dead) since Thu 2025-05-15 16:41:19 GMT; 14min ago
     Duration: 5min 22.953s
    Docs: https://httpd.apache.org/docs/2.4/
   Process: 6256 ExecStop=/usr/sbin/apachectl graceful-stop (code=exited, status=0/SUCCESS)
   Main PID: 5663 (code=exited, status=0/SUCCESS)
    CPU: 184ms

May 15 16:35:56 ibttissam-VMware-Virtual-Platform systemd[1]: Starting apache2.service - The Apache HTTP Server...
May 15 16:35:56 ibttissam-VMware-Virtual-Platform apachectl[5662]: AH00558: apache2: Could not reliably determine the se
May 15 16:35:56 ibttissam-VMware-Virtual-Platform systemd[1]: Started apache2.service - The Apache HTTP Server.
May 15 16:41:19 ibttissam-VMware-Virtual-Platform systemd[1]: Stopping apache2.service - The Apache HTTP Server...
May 15 16:41:19 ibttissam-VMware-Virtual-Platform apachectl[6258]: AH00558: apache2: Could not reliably determine the se
May 15 16:41:19 ibttissam-VMware-Virtual-Platform systemd[1]: apache2.service: Deactivated successfully.
May 15 16:41:19 ibttissam-VMware-Virtual-Platform systemd[1]: Stopped apache2.service - The Apache HTTP Server.
lines 1-16/16 (END)
```

- kill, killall : arrêt de processus

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/expo_system$ firefox &
[1] 10425
ubuntu@ubuntu-VMware-Virtual-Platform:~/expo_system$ Gtk-Message: 18:58:28.086: Not loading module "atk-bridge": The fun
ctionality is provided by GTK natively. Please try to not load it.
ps -l
F S  UID      PID      PPID    C  PRI   NI  ADDR  SZ  WCHAN  TTY          TIME CMD
0 S  1000     9541     2991    0   80    0   -   4988  do_wai pts/0        00:00:00 bash
4 S  1000    10425     9541   30   80    0   -  737109  do_pol pts/0        00:00:05 firefox
4 S  1000    10645    10425    0   80    0   -   54401  do_pol pts/0        00:00:00 Socket Process
4 S  1000    10670    10425    4   80    0   -  616642  do_pol pts/0        00:00:00 Privileged Cont
4 S  1000    10675    10425    0   80    0   -   87497  do_pol pts/0        00:00:00 RDD Process
4 S  1000   10991    10425    3   80    0   -  610198  do_pol pts/0        00:00:00 WebExtensions
4 S  1000   11168    10425    0   80    0   -   53991  do_pol pts/0        00:00:00 Utility Process
4 S  1000   11176    10425    1   80    0   -   602340  do_pol pts/0        00:00:00 Web Content
4 S  1000   11178    10425    1   80    0   -   602340  do_pol pts/0        00:00:00 Web Content
4 S  1000   11188    10425    1   80    0   -   602339  do_pol pts/0        00:00:00 Web Content
4 R  1000   11266     9541   99   80    0   -    5612  -      pts/0        00:00:00 ps
ubuntu@ubuntu-VMware-Virtual-Platform:~/expo_system$ kill 10425
ubuntu@ubuntu-VMware-Virtual-Platform:~/expo_system$ Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
ps -l
F S  UID      PID      PPID    C  PRI   NI  ADDR  SZ  WCHAN  TTY          TIME CMD
0 S  1000     9541     2991    0   80    0   -   4988  do_wai pts/0        00:00:00 bash
0 R  1000    11284     9541   99   80    0   -    5612  -      pts/0        00:00:00 ps
[1]+  Terminated                  firefox
ubuntu@ubuntu-VMware-Virtual-Platform:~/expo_system$
```

- nice, renice : priorité d'un processus

```

F S  UID      PID      PPID    C  PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
0 S   1000     9541     2991    0   80   0  -  4988 do_wai pts/0    00:00:00 bash
4 S   1000     11302    9541   45   80   0  - 2837464 do_pol pts/0    00:00:05 firefox
4 S   1000     11523    11302    0   80   0  -  54401 do_pol pts/0    00:00:00 Socket Process
4 S   1000     11544    11302    7   80   0  - 4811031 do_pol pts/0    00:00:00 Privileged Cont
4 S   1000     11549    11302    0   80   0  -  87497 do_pol pts/0    00:00:00 RDD Process
4 S   1000     11866    11302    4   80   0  -  610137 do_pol pts/0    00:00:00 WebExtensions
4 S   1000     12042    11302    1   80   0  -  53991 do_pol pts/0    00:00:00 Utility Process
4 S   1000     12050    11302    3   80   0  -  602340 do_pol pts/0    00:00:00 Web Content
4 S   1000     12052    11302    3   80   0  -  602339 do_pol pts/0    00:00:00 Web Content
4 S   1000     12082    11302    2   80   0  -  602339 do_pol pts/0    00:00:00 Web Content
4 R   1000     12142     9541   99   80   0  -   5613 - pts/0    00:00:00 ps
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ nice -n -15 firefox 5&
[2] 12172
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ nice: cannot set niceness: Permission denied
Gtk-Message: 19:08:21.638: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to
not load it.
ps -l
F S  UID      PID      PPID    C  PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
0 S   1000     9541     2991    0   80   0  -  4988 do_wai pts/0    00:00:00 bash
4 S   1000     11302    9541    4   80   0  -  738434 do_pol pts/0    00:00:07 firefox
4 S   1000     11523    11302    0   80   0  -  54401 do_pol pts/0    00:00:00 Socket Process
4 S   1000     11544    11302    0   80   0  -  616581 do_pol pts/0    00:00:01 Privileged Cont
4 S   1000     11549    11302    0   80   0  -  87562 do_pol pts/0    00:00:00 RDD Process
4 S   1000     11866    11302    0   80   0  -  610117 do_pol pts/0    00:00:00 WebExtensions
4 S   1000     12042    11302    0   80   0  -  54060 do_pol pts/0    00:00:00 Utility Process
4 S   1000     12050    11302    0   80   0  -  604981 do_pol pts/0    00:00:00 Isolated Web Co
4 S   1000     12052    11302    0   80   0  -  602339 do_pol pts/0    00:00:00 Web Content
4 S   1000     12082    11302    0   80   0  -  602339 do_pol pts/0    00:00:00 Web Content
4 S   1000     12344    11302    1   80   0  -  602340 do_pol pts/0    00:00:00 Web Content
4 R   1000     12367     9541   99   80   0  -   5613 - pts/0    00:00:00 ps
[2]+  Done                  nice -n -15 firefox 5

```

Installation de logiciels

- apt update : met à jour la liste des paquets

```

obunto@obunto-VMware-Virtual-Platform:~/expo_system$ sudo apt update
[sudo] password for obunto:
Ign:1 http://archive.ubuntu.com/ubuntu noble InRelease
Ign:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Ign:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Ign:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Ign:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Ign:1 http://archive.ubuntu.com/ubuntu noble InRelease
Ign:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Ign:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Ign:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Ign:1 http://archive.ubuntu.com/ubuntu noble InRelease
Ign:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Ign:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,090 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [232 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [161 kB]

```


- apt install nom : installe un paquet logiciel

```
obunto@obunto-VMware-Virtual-Platform:~/expo_system$ sudo apt install curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  curl
0 upgraded, 1 newly installed, 0 to remove and 116 not upgraded.
Need to get 226 kB of archives.
After this operation, 534 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 curl amd64 8.5.0-2ubuntu10.6 [226 kB]
Fetched 226 kB in 5s (45.4 kB/s)
Selecting previously unselected package curl.
(Reading database ... 196597 files and directories currently installed.)
Preparing to unpack .../curl_8.5.0-2ubuntu10.6_amd64.deb ...
Unpacking curl (8.5.0-2ubuntu10.6) ...
Setting up curl (8.5.0-2ubuntu10.6) ...
Processing triggers for man-db (2.12.0-4build2) ...
```

Erreurs rencontrées et solutions

- Problème de connexion réseau dans la VM → Solution : activation du mode “bridged” dans VMware
- Accès refusé à certaines commandes → Solution : utilisation de sudo
- Erreur dans l’utilisation de chmod → Solution : vérification et utilisation correcte des permissions numériques
- Utilisateur non ajouté au groupe → Solution : utilisation correcte de usermod -aG

Conclusion

Ce projet nous a permis d’apprendre à utiliser les commandes essentielles de Linux, à gérer des utilisateurs, des fichiers, des services et à surveiller le fonctionnement du système. Il constitue une bonne base pour progresser en administration système et comprendre le fonctionnement global d’un serveur Linux.